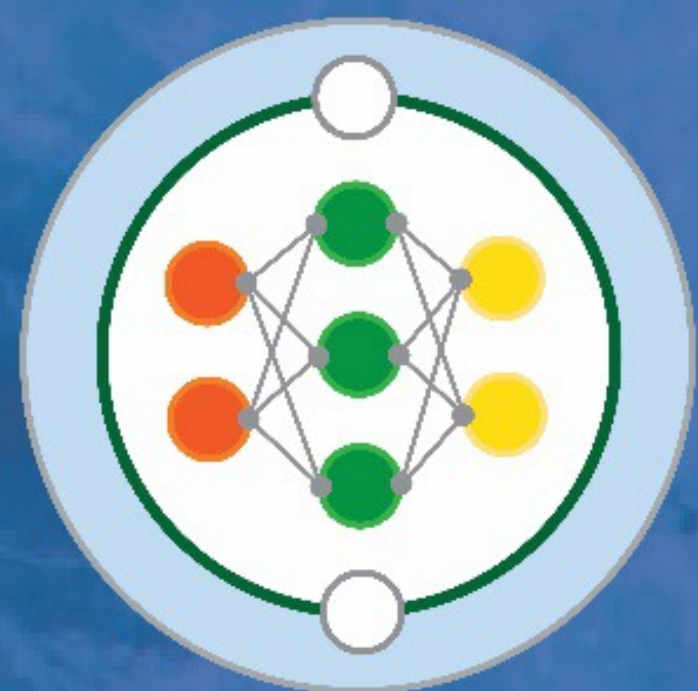




m2lines.github.io

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Calibration of a neural network ocean closure for improved mean state and variability

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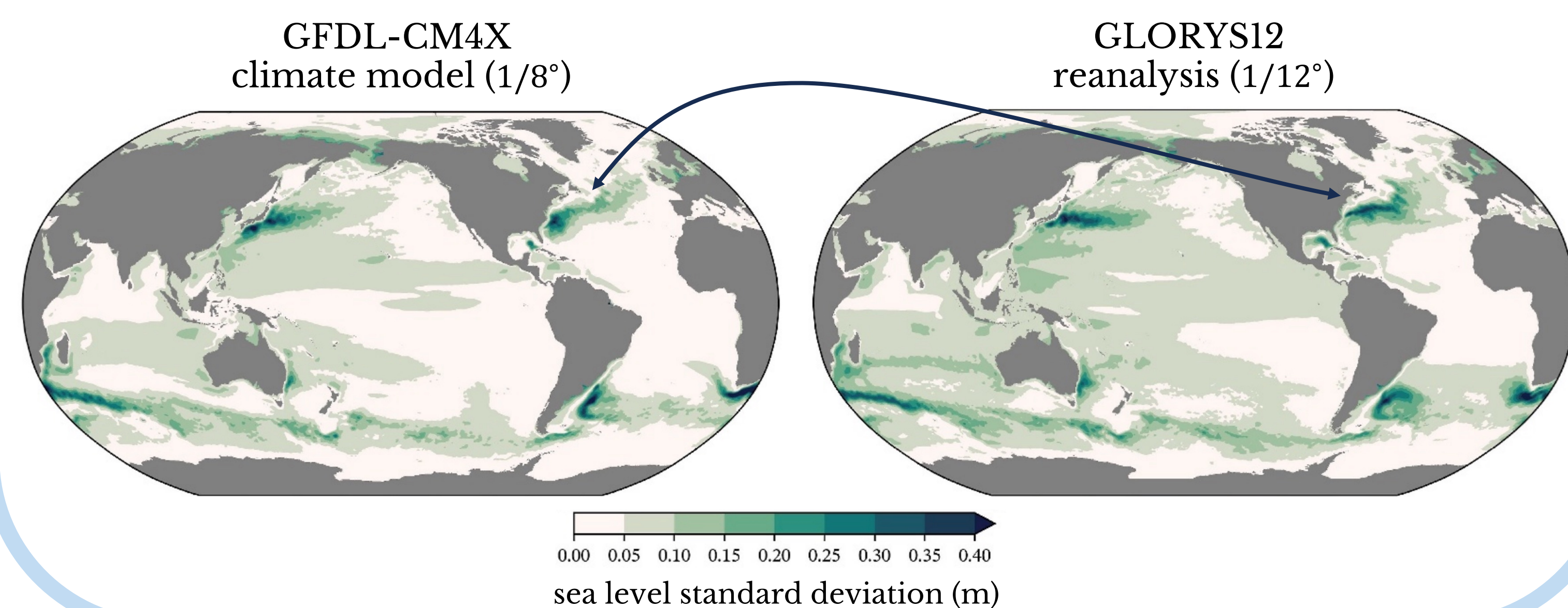
(a) Courant Institute, New York University, (b) Program in Atmospheric and Oceanic Sciences, Princeton



SCHMIDT FUTURES

Motivation

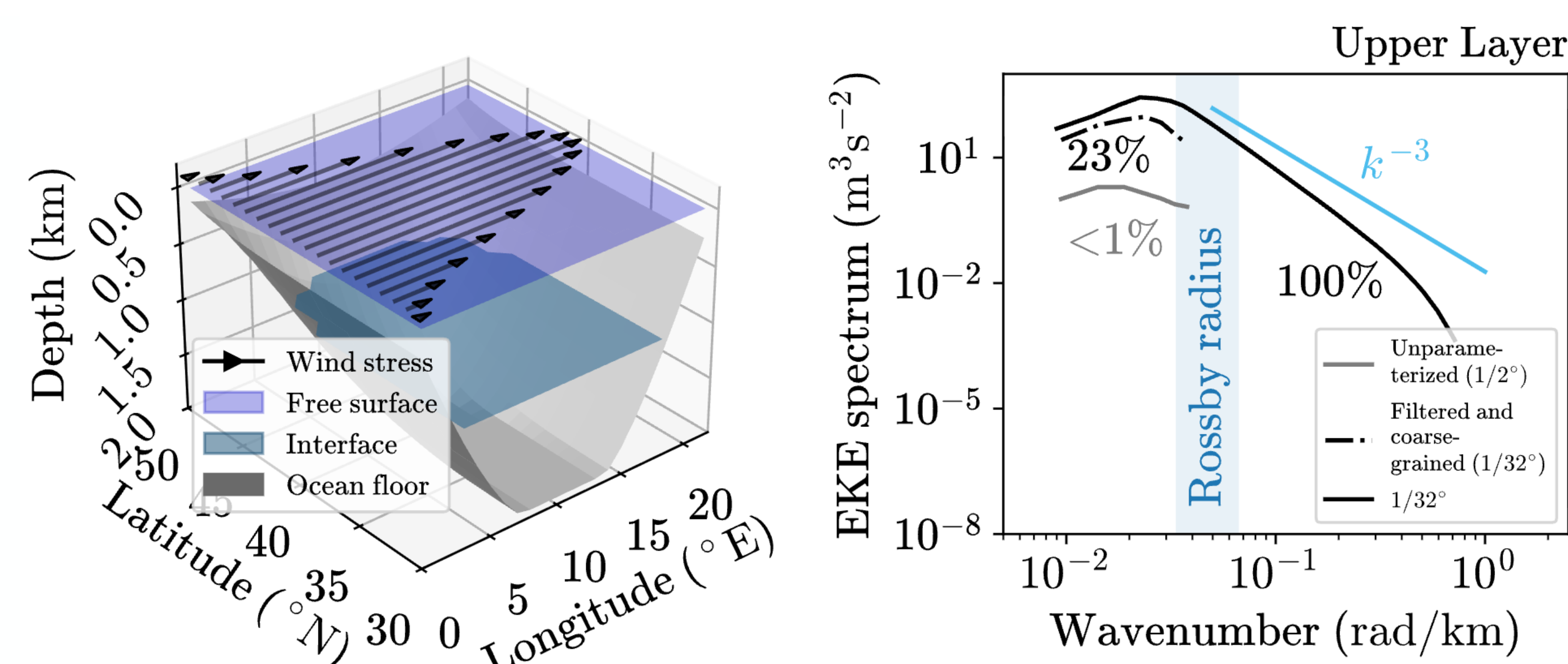
Ocean components of climate models demonstrate biases in the **mean state** and **variability** even at relatively high resolutions ($1/4^\circ - 1/8^\circ$)
Griffies, Stephen M., et al. "The GFDL-CM4X climate model hierarchy, Part I: Model description and thermal properties." *Journal of Advances in Modeling Earth Systems* (2025)



Challenging idealized setup

GFDL MOM6 2-layer wind-driven Double Gyre (DG) simulation:

- ❖ Less than 1% of eddy kinetic energy (EKE) is reproduced in unparameterized simulation at resolution $1/2^\circ$
- ❖ Only 23% of EKE is resolved in filtered and coarsegrained data



Biases are addressed by invoking neural network (eANN) parameterization of **mesoscale eddies** by Perezhogin, Adcroft & Zanna, GRL (2025):

$$\partial_t \mathbf{u} = \dots + \nabla \cdot \mathbf{T}$$

$$\mathbf{T}(\mathbf{X}, \Delta) = \gamma \Delta^2 \|\mathbf{X}\|_2^2 f_\phi(\mathbf{X}/\|\mathbf{X}\|_2)$$

$$\mathbf{X} = \begin{pmatrix} [\bar{\sigma}_S] \uparrow 9 \\ [\bar{\sigma}_T] \uparrow 9 \\ [\bar{\omega}] \uparrow 9 \end{pmatrix} \in \mathbb{R}^{27} \quad \begin{array}{ll} \bar{\sigma}_S = \partial_y \bar{u} + \partial_x \bar{v} & \text{— shearing strain,} \\ \bar{\sigma}_T = \partial_x \bar{u} - \partial_y \bar{v} & \text{— horizontal tension/stretch,} \\ \bar{\omega} = \partial_x \bar{v} - \partial_y \bar{u} & \text{— relative vorticity.} \end{array}$$

Improving mean state and variability as a calibration problem

We consider optimization problem with loss function:

$$\mathcal{L}(\theta) = \|y - \mathcal{G}(\theta)\|_2^2$$

y is the **observational vector**, i.e. statistics of filtered and coarsegrained high-resolution simulation

θ is the **parameter vector**, i.e. weights and biases of the last layer of the neural network

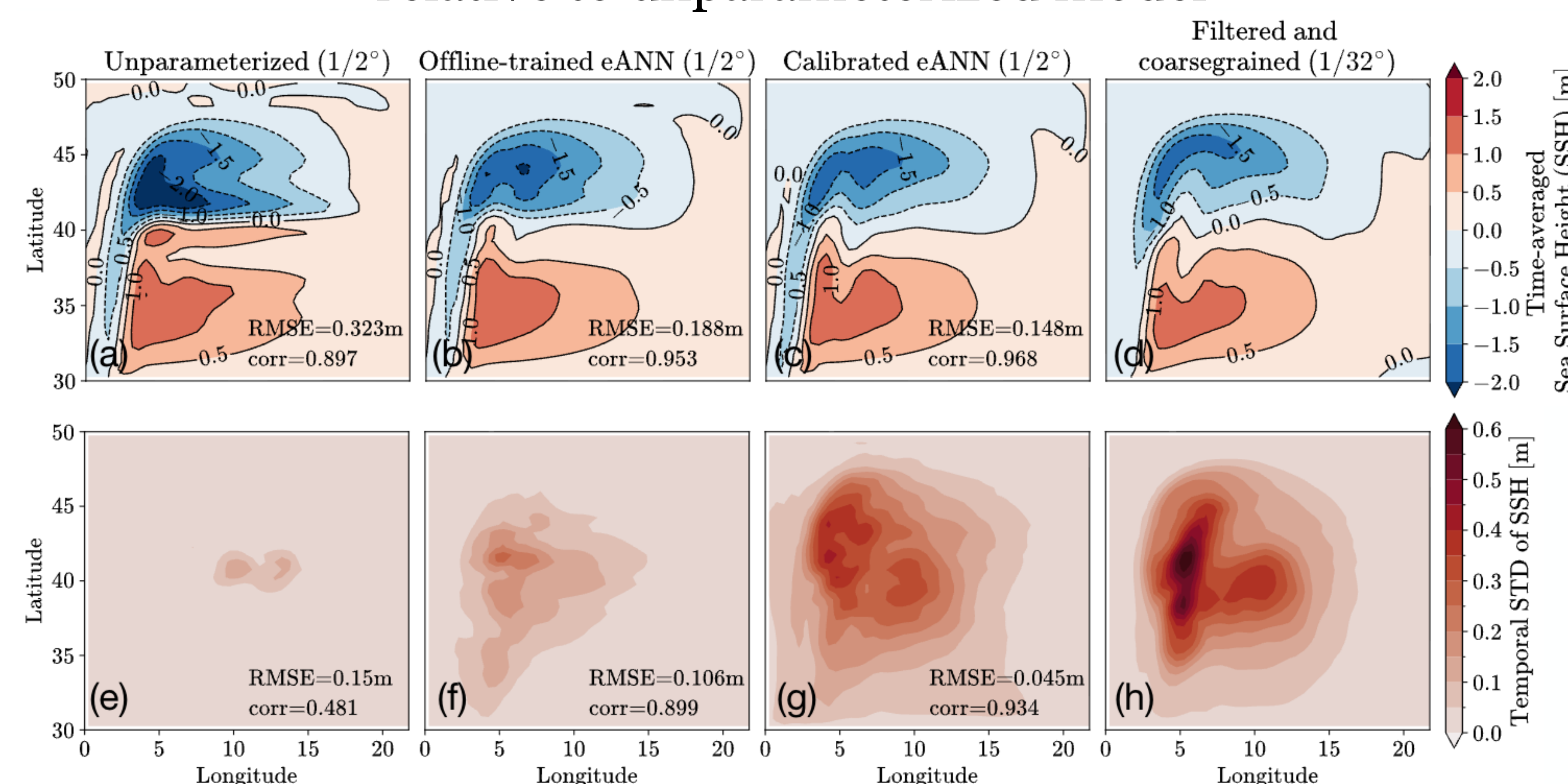
$\mathcal{G}(\theta)$ is the **forward model**, i.e. statistics of coarse parameterized simulation

Statistics are defined as follows:

$$y = \begin{bmatrix} \bar{\eta}^t \\ \sqrt{\eta'^2}^t \end{bmatrix} \quad \begin{array}{l} \bar{\eta}^t \text{ is 10-years averaged fluid interfaces} \\ \sqrt{\eta'^2}^t \text{ is the standard deviation of fluid interfaces} \end{array}$$

Gradient-free optimization with Ensemble Kalman Inversion (EKI[†])

Calibrated parameterization reduces errors in mean state and variability by factors of **2.2** and **3.3** relative to unparameterized model



Ensemble-mean update:

$$\bar{\theta}_{j+1} = \bar{\theta}_j + \Theta_j (I + G_j^T G_j)^{-1} G_j^T (y - \bar{g}_j)$$

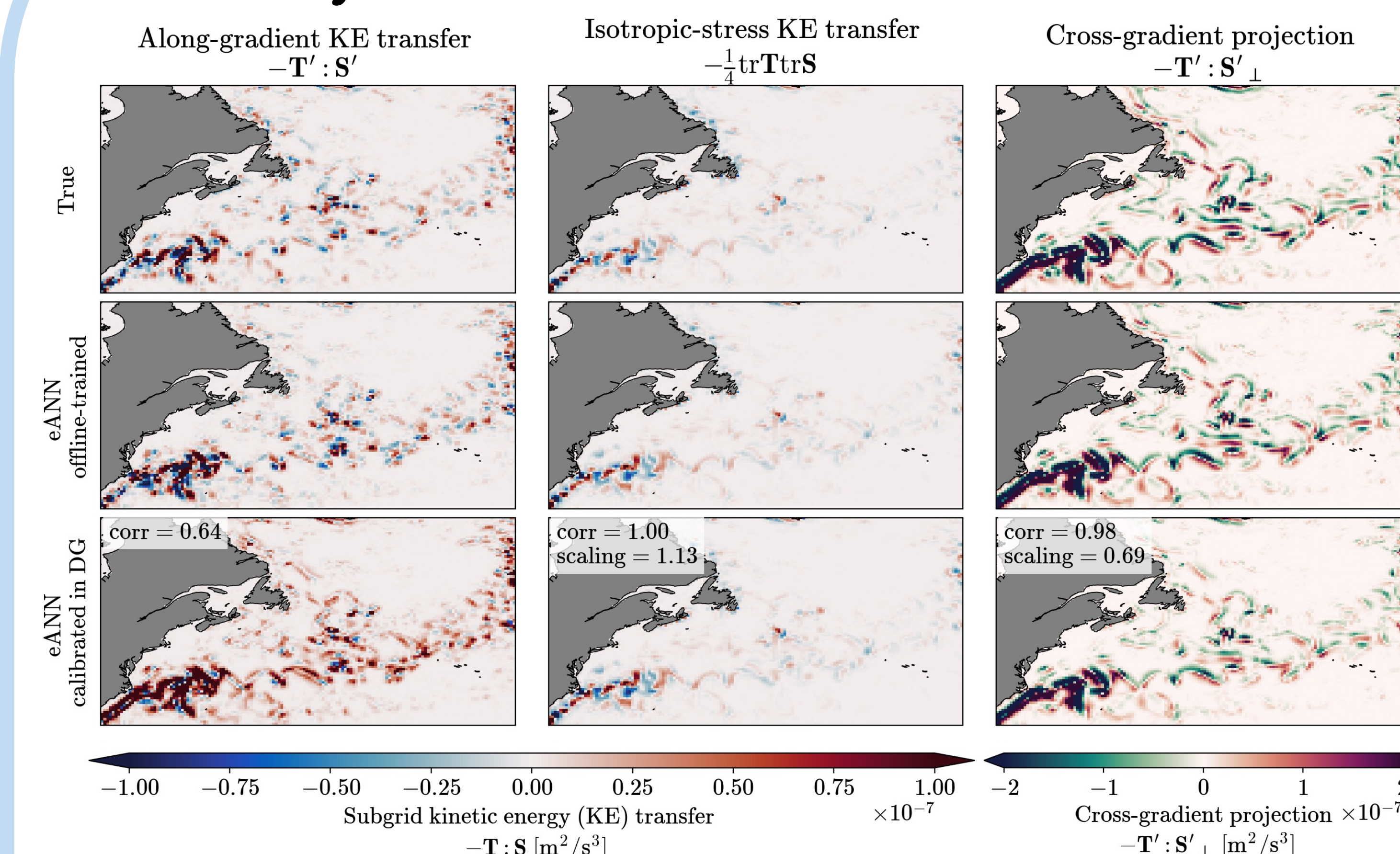
Ensemble-perturbations update:

$$\Theta_{j+1} = \Theta_j (I + G_j^T G_j)^{-1/2}$$

$\bar{\theta}_j$ is mean parameter vector
 $\bar{g}_j$ is mean forward model output
 Θ_j, G_j are ensemble perturbations

[†]Iglesias, Marco A., Kody JH Law, and Andrew M. Stuart. "Ensemble Kalman methods for inverse problems." *Inverse Problems* (2013)
[†]Huang, Daniel Zhengyu, et al. "Efficient derivative-free Bayesian inference for large-scale inverse problems." *Inverse Problems* (2022)
[†]Gjini, Rebecca, et al. "The Ensemble Kalman Inversion Race." *arXiv* (2023)

Physical effect of calibration



Calibration **modifies** offline-trained parameterization as follows:

- ❖ Along-gradient fluxes are **skewed** towards **stronger backscatter** (compensation of numerical viscosity)
- ❖ Isotropic stress is **unchanged** (as it is less important)
- ❖ Cross-gradient fluxes are **attenuated** (adjust effective resolution)

Conclusions

- ❖ **Mean state and variability can be improved** by calibrating data-driven parameterization of **mesoscale eddies**
- ❖ Calibrated parameterization learns **new spatial patterns** which cannot be obtained by simple retuning of the scaling coefficient
- ❖ EKI allows to solve optimization problem for **noisy non-differentiable forward models**
- ❖ See our paper for scaling to **more challenging setup** (NeverWorld2) and a method to **avoid integration to statistical equilibrium**

Poster PDF



Paper PDF

